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This box is for the Examiner only.

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 10 | 9 | 13 | 6 | 7 | 15 | 60 |
| Score: | | | | | | | |

# Questions

1. **[Measuring the Economy]**

Suppose that an economy consists of only two types of products: beer and brezels. Sales and price data for these two products for two different years are as shown below:

| Year | No. of beers sold | Price per beer | No. of brezels | Price per brezel |
|---|---|---|---|---|
| 2018 | 1,000,000 | €3 | 2,000,000 | €1 |
| 2019 | 1,000,000 | €2 | 2,000,000 | €2 |

(a) (4 points) Assuming that beer and brezels are final goods, calculate nominal GDP in 2018 and in 2019.

(b) (2 points) Calculate the percentage change in **nominal GDP** between 2018 and 2019.

(c) (2 points) Calculate the consumer price index (CPI) in 2018 and 2019, using 2018 as the base year.

(d) (1 point) Calculate the percentage change in the consumer price index between 2018 to 2019.

(e) (1 point) Calculate the percentage change in **real GDP** between 2018 and 2019.

2. (9 points) **[Consumer Price Index (CPI)]**

The consumer price index (CPI) is an accurate measure of the selected goods that aim to be a representative bundle of goods, but it is not a perfect measure of the cost of living. Why? Explain briefly the three sources of bias which we discussed in class.

3. **[Unemployment]** If nominal wages are downward rigid and above the market clearing level, involuntary unemployment exists.

   (a) (4 points) Sketch this scenario in a two-way graph. Please, don't forget to label axes and lines, respectively.

   (b) (9 points) There are several rational explanations why wages are downward rigid and remain above the market clearing level. Explain the three sources that we discussed in class.

4. **[European Central Bank]**

(a) (2 points) State the primary objective of the European Central Bank.

(b) (4 points) One of the most powerful instruments of central banks to influence prices are policy rates. In the course, we discussed the limited power of policy rates to stimulate investments and increase prices, respectively. Explain why the power of a further decrease in policy rates to stimulate an economy is limited once the rate is negative, i.e., below zero.

5. (7 points) [**International Investments**] In the lecture, we discussed three components to see how the value of an investment changes over time, when it was invested in an asset abroad. Describe the three components and write down the formula that allows to calculate the rate of return of an international investment.

6. **[IS-LM Model]**

The IS relation is defined by the equation

$$Y = C(Y - T) + I(Y, i) + G.$$

The LM relation is defined by the equation

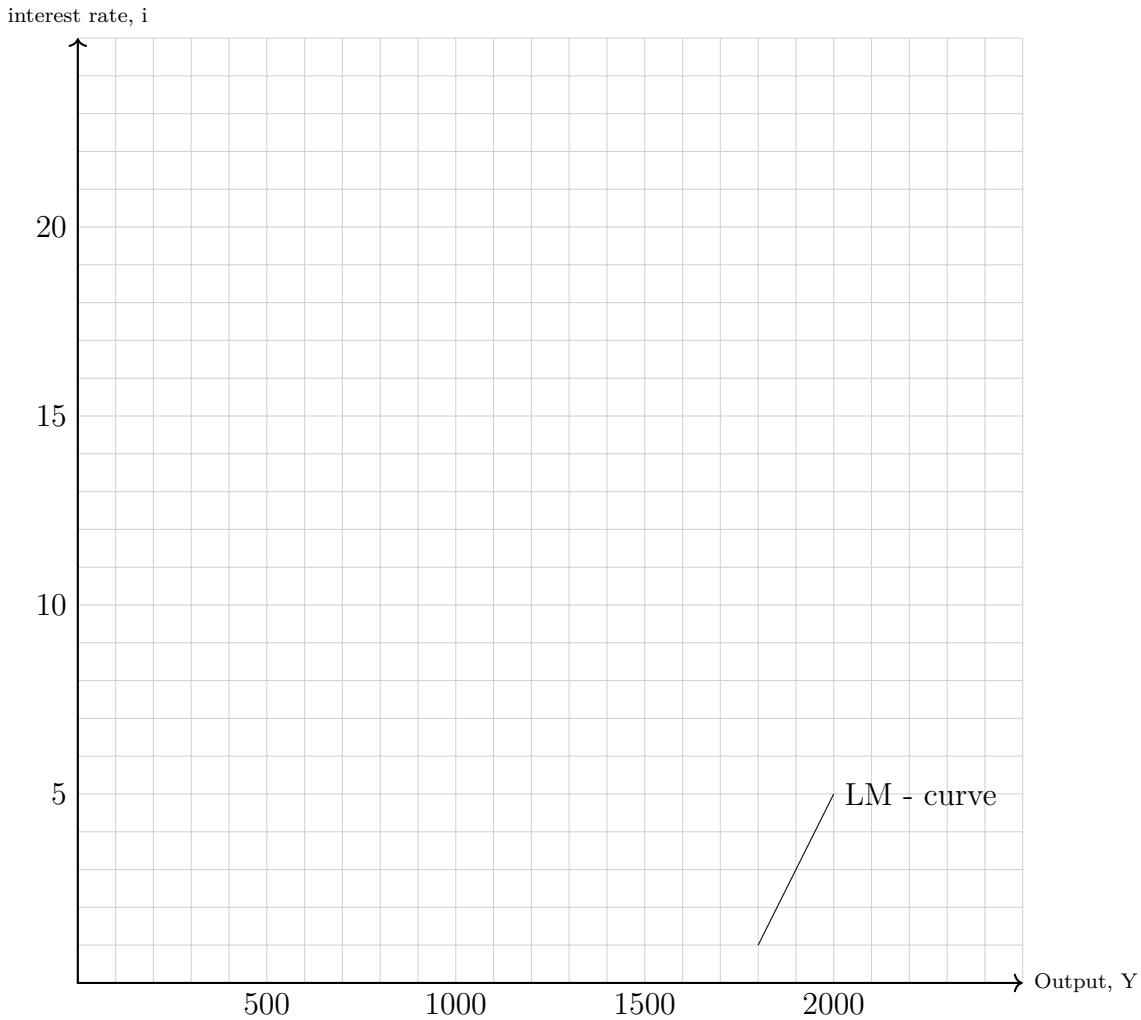
$$\frac{M}{P} = YL(i).$$

Consider an economy described by the following equations:

$$\begin{aligned} Y &= C + I + G \\ C &= 400 + 0.25(Y - T)) \\ I &= 300 + 0.25Y - 50i \\ G &= 500 \\ T &= 400 \end{aligned}$$

(a) (5 points) Derive the IS curve.

(b) (4 points) Draw the IS curve as derived in (a) into the plot below. Look on the intersection of the IS-curve with the LM-curve and state the equilibrium output and interest rate.




(c) (3 points) Explain the Taylor-rule as an alternative to the LM-curve.

(d) (3 points) Fill in the gaps of the following text:

- A monetary expansion leads to _____ (hint: *higher or lower*) output and a _____ (hint: *higher or lower*) interest rate.
- An increase in taxes shifts the IS curve to the _____ (hint: *left or right*) and leads to a _____ (hint: *increase or decrease*) in the equilibrium level of output and the equilibrium interest rate.
- An increase in government spending, $G$, shifts the IS curve to the _____ (hint: *left or right*) and leads to a _____ (hint: *increase or decrease*) in the equilibrium level of output and the equilibrium interest rate.

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This box is for the Examiner only.

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 15 | 14 | 10 | 10 | 7 | 4 | 60 |
| Score: | | | | | | | |

# Questions

1. **[IS-LM Model]**

The IS relation is defined by the equation

$$Y = C(Y - T) + I(Y, i) + G.$$

The LM relation is defined by the equation

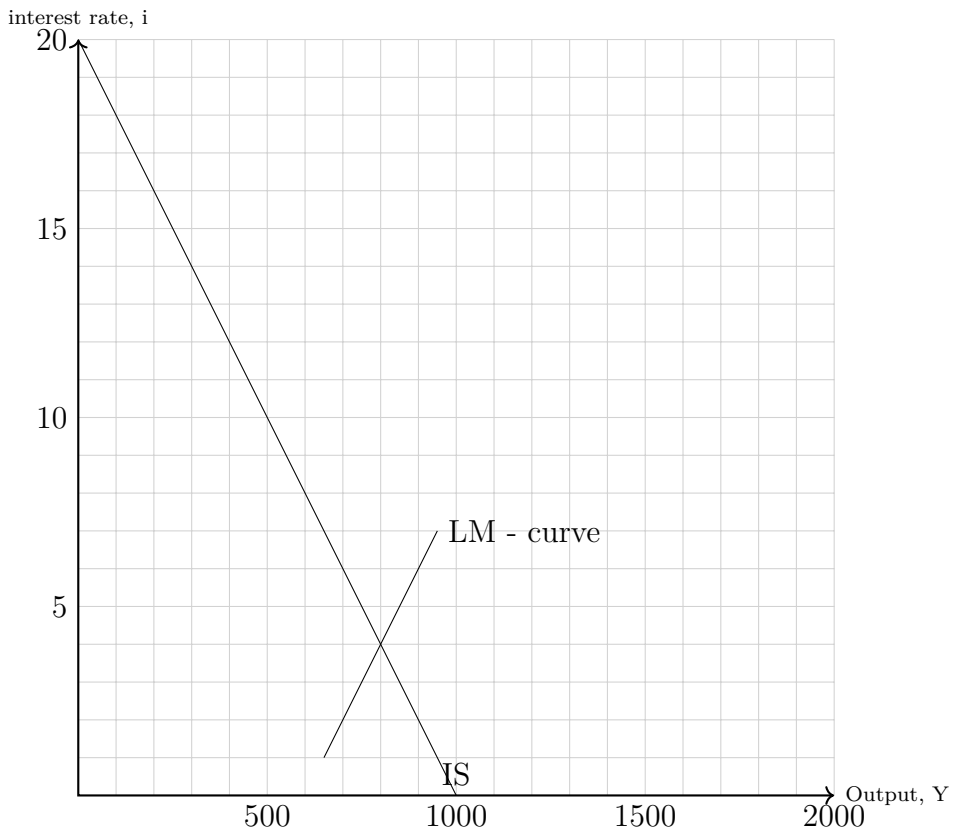
$$\frac{M}{P} = YL(i).$$

Consider an economy described by the following equations:

$$\begin{aligned} Y &= C + I + G \\ C &= 400 + 0.25(Y - T)) \\ I &= 300 + 0.25Y - 50i \\ G &= 50 \\ T &= 400 \end{aligned}$$

(a) (6 points) Derive the IS curve.

(b) (6 points) Draw the IS curve as derived in (a) into the plot below. Look on the intersection of the IS-curve with the LM-curve and state the equilibrium output and interest rate.



(c) (3 points) Fill in the gaps of the following text:

- A monetary expansion leads to _____ (hint: *higher or lower*) output and a _____ (hint: *higher or lower*) interest rate.
- An increase in taxes shifts the IS curve to the _____ (hint: *left or right*) and leads to a _____ (hint: *increase or decrease*) in the equilibrium level of output and the equilibrium interest rate.
- An increase in government spending, $G$, shifts the IS curve to the _____ (hint: *left or right*) and leads to a _____ (hint: *increase or decrease*) in the equilibrium level of output and the equilibrium interest rate.

2. **[Consumer Price Index (CPI)]**

(a) (5 points) Explain briefly the five steps to calculate the ***Consumer Price Index***.

(b) (9 points) The consumer price index (CPI) is an accurate measure of the selected goods that aim to be a representative bundle of goods, but it is not a perfect measure of the cost of living. Why? Explain three sources of bias.

3. **[European Central Bank]**

(a) (2 points) State the primary objective of the European Central Bank.

(b) (4 points) One of the most powerful instruments of central banks to influence prices are policy rates. In the lecture, we described four ways how the interest/policy rate policy of a Central bank can increase investments. Explain two of them.

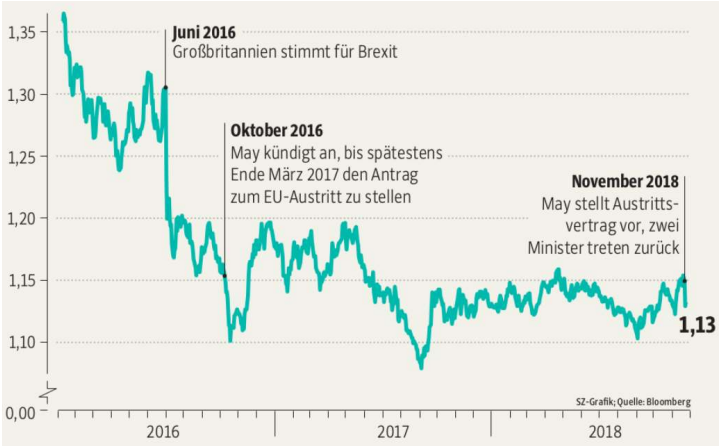
(c) (4 points) Explain why the power of a further decrease in policy rates to stimulate an economy is limited once the rate is negative, i.e., below zero.

4. **[Unemployment]**

(a) (4 points) If nominal wages are downward rigid and above the market clearing level, involuntary unemployment exists. Sketch this scenario in a two-way graph. Please, don't forget to label axes and lines, respectively.

(b) (6 points) There are several rational explanations why wages are downward rigid and remain above the market clearing level. Explain the two sources that we discussed in class.

5. [**Foreign Exchange Market**]




(a) (3 points) The time series above shows the exchange rate of the British Pound Sterling (£). The y-axis is not labeled.

- State how the exchange rate is denoted here.
- Has the Pound appreciated or depreciated against the Euro since June 2015?

(b) (2 points) Suppose you have €100 and you want to exchange it into Swiss Franc. Calculate how much £ you receive for your €100 when the exchange rate is 0.92 denoted in Swiss Franc to Euro.

(c) (2 points) The Swiss central bank unpegged the Franc from the Euro in 2015. Referring to the interest parity condition, name the two major economic consequences without going into details.

6. (4 points) [**Measuring the Economy**] Define the disposable income and state the main difference to the gross domestic product (GDP).

21-01_macro_mock.pdf

This box is for the examiner only.

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | Total |
|---|---|---|---|---|---|---|---|---|---|
| Points: | 14 | 10 | 8 | 15 | 10 | 10 | 7 | 4 | 78 |
| Score: | | | | | | | | | |

1. **[Measuring the Economy]**

An economy produces three goods: cars, computers, and oranges. Quantities and prices per unit for years 2005 and 2006 are as follows:

| | 2005 | | 2006 | |
|---|---|---|---|---|
| | Quantity | Price | Quantity | Price |
| Cars | 10 | $2000 | 12 | $3000 |
| Computers | 4 | $1000 | 6 | $500 |
| Oranges | 1000 | $1 | 1000 | $1 |

(a) (2 points) What is nominal GDP in 2005 and in 2006? By what percentage does nominal GDP change from 2005 to 2006?

**Solution:** 2005 GDP: 10($2,000)+4($1,000)+1000($1)=$25,000
2006 GDP: 12($3,000)+6($500)+1000($1)=$40,000
Nominal GDP has increased by 60%.

(b) (2 points) Using the prices for 2005 as the set of common prices, what is real GDP in 2005 and in 2006? By what percentage does real GDP change from 2005 to 2006?

**Solution:** 2005 real (2005) GDP: $25,000
2006 real (2005) GDP: 12($2,000)+6($1,000)+1000($1)=$31,000
Real (2006) GDP has increased by 24%.

(c) (2 points) Using the prices for 2006 as the set of common prices, what is real GDP in 2005 and in 2006? By what percentage does real GDP change from 2005 to 2006?

**Solution:** 2006 real (2006) GDP: 10($3,000)+4($500)+1,000($1)=$33,000
2007 real (2006) GDP: $40,000.
Real (2006) GDP has increased by 21.2%.

(d) (2 points) Why are the two output growth rates constructed in (b) and (c) different? Which one is correct? Explain your answer.

**Solution:** The answers measure real GDP growth in different units. Neither answer is incorrect, just as measurement in inches is not more or less correct than measurement in centimeters.

(e) (2 points) Use the prices for 2005 as the set of common prices to compute real GDP in 2005 and in 2006. Compute the GDP deflator for 2005 and for 2006 and compute the rate of inflation from 2005 to 2006.

> **Solution:** 2006 base year:
> Deflator(2006)=1;
> Deflator(2007)=$40,000/31$,000=1.29
> Inflation=29

(f) (2 points) Use the prices for 2006 as the set of common prices to compute real GDP in 2005 and in 2006. Compute the GDP deflator for 2005 and for 2006 and compute the rate of inflation from 2005 to 2006.

> **Solution:** 2007 base year:
> Deflator(2005)=$25,000/33$,000=0.76;
> Deflator(2006)=1
> Inflation=(1-0.76)/0.76=.32=32%

(g) (2 points) Why are the two rates of inflation different? Which one is correct? Explain your answer.

> **Solution:** Analogous to (d)

2. (10 points) **[Measuring the Economy]** Explain in detail the five steps to calculate the Consumer Price Index.

3. (8 points) **[Measuring the Economy]** Discuss why the nominal GDP is a bad predictor of well-beeing across countries. Can you imagine better alternatives?

4. **[Economies in the Short-Run]**

   (a) (6 points) Explain what we have discussed in class as the *Reverse Multiplier Effect*.

   (b) (4 points) Discuss briefly the level of unemployment when market are *perfect*.

   (c) (5 points) Explain the differences between *structural unemployment* and *frictional unemployment*.

5. **[European Central Bank]**

   (a) (2 points) State the primary objective of the European Central Bank.

   (b) (4 points) One of the most powerful instruments of central banks to influence prices are policy rates. In the lecture, we described four ways how the interest/policy rate policy of a Central bank can increase investments. Explain two of them.

(c) (4 points) Explain why the power of a further decrease in policy rates to stimulate an economy is limited once the rate is negative, i.e., below zero.

6. **[Unemployment]**

(a) (4 points) If nominal wages are downward rigid and above the market clearing level, involuntary unemployment exists. Sketch this scenario in a two-way graph. Please, don't forget to label axes and lines, respectively.

(b) (6 points) There are several rational explanations why wages are downward rigid and remain above the market clearing level. Explain the two sources that we discussed in class.

7. **[Foreign Exchange Market]**

(a) (3 points) The time series above shows the exchange rate of the British Pound Sterling (£). The y-axis is not labeled.

- State how the exchange rate is denoted here.
- Has the Pound appreciated or depreciated against the Euro since June 2015?

(b) (2 points) Suppose you have €100 and you want to exchange it into Swiss Franc. Calculate how much Swiss Franc you receive for your €100 when the exchange rate is 0.92 denoted in Swiss Franc to Euro.

(c) (2 points) The Swiss central bank unpegged the Franc from the Euro in 2015. Referring to the interest parity condition, name the two major economic consequences without going into details.

8. (4 points) **[Measuring the Economy]** Define the disposable income and state the main difference to the gross domestic product (GDP).

*End of exam*

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**This box is for the examiner only.**

| Question: | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Points: | 6 | 11 | 18 | 9 | 16 | 60 |
| Score: | | | | | | |

# Questions

1. (6 points) [**Measuring the Economy (Part A)**]

   Discuss why the GDP per capita is only an imperfect measure for welfare and well-being.

2. **[Measuring the Economy (Part B)]**

Consider the following table of price and quantities produced for a small economy.

| Year | Masks | | Toilete Paper | | Cinema Tickets | | Wine | | Grapes | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Price | Quantity | Price | Quantity | Price | Quantity | Price | Quantity | Price | Quantity |
| 2019 | €1 | 200 | €2 | 100 | 10 | 10 | 3 | 30 | 1 | 15 |
| 2020 | €1 | 1000 | €2 | 100 | 10 | 0 | 2 | 40 | 1 | 20 |

(a) (4 points) Assume that all grapes in this economy are used to make wine. Compute nominal GDP for 2019 and 2020.

(b) (2 points) Calculate the percentage change in nominal GDP between 2019 and 2020.

(c) (2 points) Continue to assume that all grapes are used to make wine. Using 2019 as the base year, compute real GDP in 2019 and 2020.

(d) (2 points) Calculate the consumer price index (CPI) in 2019 and 2020, using 2019 as the base year.

(e) (1 point) What was the inflation rate in this economy between 2019 and 2020? Express your answer as a percentage.

3. **[Solow Model]**

Consider the Solow model of economic growth as introduced in the lecture. Assume the production function is

$$Y = K^{1/2}L^{1/2}$$

where $Y$ is output, $K$ is the capital stock, and $L$ is the labor force. The labor force (assumed equal to the population) grows at a constant rate $n$. The capital stock depreciates at a constant rate $\delta$. There is no exogenous technological progress ($g = 0.02$). The saving rate is $s$.

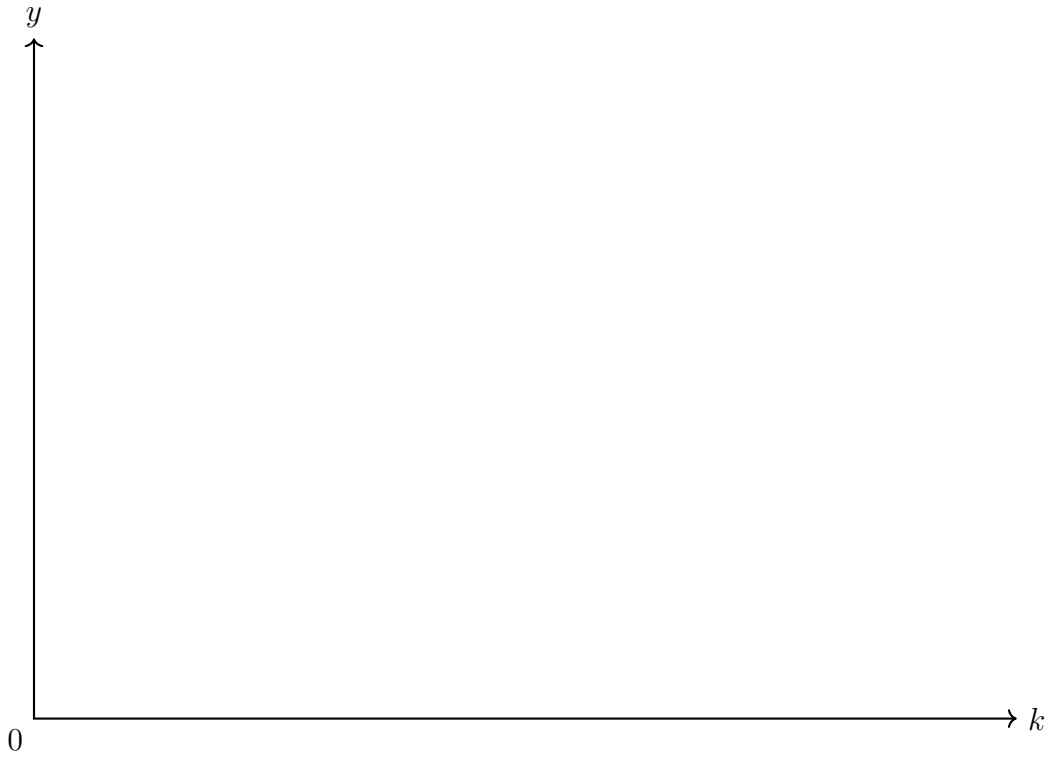
(a) (4 points) Show that the production function satisfies the assumption of the Solow model.

(b) (3 points) Write the production function in the intensive form, i.e, in output per effective unit of labor terms.

(c) (8 points) The dynamics of the capital stock per person are described by the equation

$$\dot{k} = sf(k) - (\delta - n)k.$$

Show how the steady-state stock of capital per worker is found using the diagram below and explain why the economy will converge to this point in the long run. Moreover, sketch in the figure the overall output per worker function, $f(k)$, and show how much of the overall output per worker goes to consumption.



(d) (1 point) Can a country increase its speed of economic growth in the short run by an increase of its saving rate?

- ◯ YES
- ◯ NO

(e) (1 point) Can a country increase its speed of economic growth in the long run by an increase of its saving rate?

- ◯ NO
- ◯ YES

(f) (1 point) What is the speed of economic growth for the countries of this model in the long run.

4. (9 points) **[Unemployment]**

In a *perfect labor market* no unemployment exists because wages adjust so that the quantity of labor supplied and the quantity of labor demanded would be equal. When the wage is above the equilibrium level the quantity of labor supplied exceeds the quantity of labor demanded and hence unemployment exists. Explain some reasons why the wage does not adjust downwards so that labor markets do not clear in equilibrium.

5. **[IS-LM Model]**

Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
Y &= C + I + G \\
C &= 400 + 0.25(Y - T) \\
I &= 300 + 0.25Y - 50i \\
G &= 500 \\
T &= 400 \\
\left(\frac{M}{P}\right)^d &= 2Y - 400i \\
\frac{M}{P} &= 600
\end{aligned}
$$

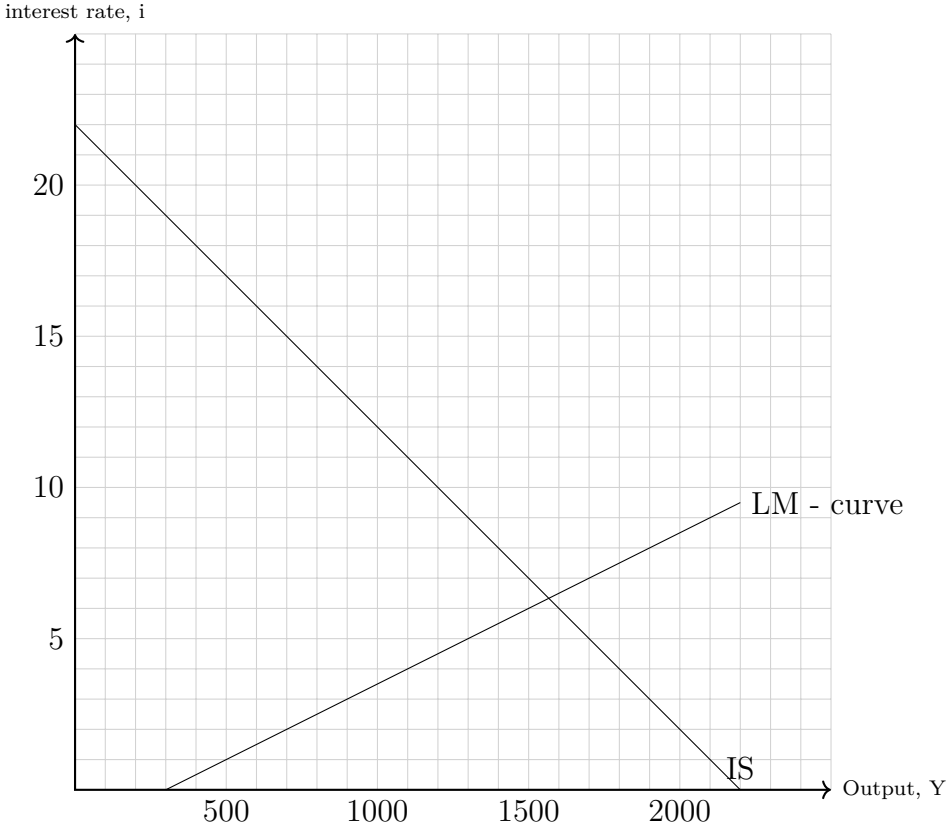
(a) (6 points) Derive the IS relation.

(b) (3 points) Derive the LM relation.

(c) (3 points) Solve for equilibrium real output.

(d) (2 points) Use the plot below to sketch graphically the impact of an increase in government spending on the IS-LM relations. Show the new equilibrium.

(e) (2 points) Use the plot below to sketch graphically the impact of a monetary expansion on the IS-LM relations. Show the new equilibrium.



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This box is for the examiner only.

| Question: | 1 | 2 | 3 | 4 | Total |
|---|---|---|---|---|---|
| Points: | 10 | 10 | 19 | 21 | 60 |
| Score: | | | | | |

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# Questions

1. **[Measuring the Economy]**

   (a) (5 points) The *GDP per capita* is a monetary measure of the market value of all the final goods and services produced in one year. Discuss why the *GDP per capita* is an imperfect measure for welfare and human well-being.

   (b) (5 points) State which of the following are a part of the *GDP* and which are not part of the *GDP*.

   A) Social security payments received by a retired factory worker.
   ◯ Part of GDP ◯ Not Part of GDP

   B) Unpaid services of a family member in painting the family home.
   ◯ Part of GDP ◯ Not Part of GDP

   C) The income of a butcher.
   ◯ Part of GDP ◯ Not Part of GDP

   D) The monthly allowance a college student receives from home.
   ◯ Part of GDP ◯ Not Part of GDP

   E) Rent received on an apartment.
   ◯ Part of GDP ◯ Not Part of GDP

   F) A €2 million increase in business inventories.
   ◯ Part of GDP ◯ Not Part of GDP

   G) Interest on a BMW bond.
   ◯ Part of GDP ◯ Not Part of GDP

   H) The purchase of an insurance policy.
   ◯ Part of GDP ◯ Not Part of GDP

2. **[Interest Rate Policy]**

   (a) (6 points) Explain briefly four ways and channels, respectively, on how interest rate policy can have a positive effect on investments.

(b) (4 points) Explain briefly why interest rate policy is less efficient in spurring investments when policy rates set by a central bank are already negative.

3. **[Solow Model]**

Consider the Solow model of economic growth as introduced in the lecture. Assume the production function is

$$Y = K^{1/2}L^{1/2}$$

where $Y$ is output, $K$ is the capital stock, and $L$ is the labor force. The labor force (assumed equal to the population) grows at a constant rate $n$. The capital stock depreciates at a constant rate $\delta$. There is no exogenous technological progress ($g = 0.02$). The saving rate is $s$.

(a) (5 points) Show that the production function satisfies the assumption of the Solow model.

(b) (3 points) Write the production function in the intensive form, i.e, in output per effective unit of labor terms.

(c) (8 points) The dynamics of the capital stock per person are described by the equation

$$\dot{k} = sf(k) - (\delta + n)k.$$

Sketch how the steady-state stock of capital per worker is found using a $y$-$k$ diagram and explain why the economy will converge to this point in the long run. Moreover, sketch in the figure the overall output per worker function, $f(k)$, and show how much of the overall output per worker goes to consumption.

(d) (1 point) Can a country increase its speed of economic growth in the short run by an increase of its saving rate?

◯ YES

◯ NO

(e) (1 point) Can a country increase its speed of economic growth in the long run by an increase of its saving rate?

◯ NO

◯ YES

(f) (1 point) What is the speed of economic growth for the countries of this model in the long run.

4. **[IS-LM Model]**

Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
Y &= C + I + G \\
C &= 300 + 0.25(Y - T) \\
I &= 400 + 0.25Y - 50i \\
G &= 200 \\
T &= 200 \\
\left(\frac{M}{P}\right)^d &= 2Y - 200i \\
\frac{M}{P} &= 600
\end{aligned}
$$

(a) (8 points) Derive the IS relation.

(b) (3 points) Derive the LM relation.

(c) (4 points) Solve for equilibrium output, $Y^*$, and interest rate, $i^*$.

(d) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if government would –all else equal– reduce the tax?

- ◯ $i^*$ would increase
- ◯ $i^*$ would decrease
- ◯ $Y^*$ would increase
- ◯ $Y^*$ would decrease

(e) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if government would –all else equal– reduce the government spendings $G$?

- ◯ $i^*$ would increase
- ◯ $i^*$ would decrease
- ◯ $Y^*$ would increase
- ◯ $Y^*$ would decrease

(f) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if the central bank would –all else equal– increase money supply, $\frac{M}{P}$?

- ◯ $i^*$ would increase
- ◯ $i^*$ would decrease
- ◯ $Y^*$ would increase
- ◯ $Y^*$ would decrease

21-04_macro_solution.pdf

This box is for the examiner only.

| Question: | 1 | 2 | 3 | 4 | Total |
|---|---|---|---|---|---|
| Points: | 10 | 10 | 19 | 21 | 60 |
| Score: | | | | | |

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# Questions

1. **[Measuring the Economy]**

   (a) (5 points) The *GDP per capita* is a monetary measure of the market value of all the final goods and services produced in one year. Discuss why the *GDP per capita* is an imperfect measure for welfare and human well-being.

   **Solution:** The *GDP per capita* is an imperfect measure because goods and services that actually indicate a bad well-being and catastrophic circumstances (guns, tanks, poisson, medicine, masks, drugs, etc.) and certain goods and services remain outside the measure such as non-market (private) production and services. It is a market oriented measure not a measure that is directly linked to the happiness of the people.

   Moreover, *GDP per capita* does not account for the degree of income inequality in a society. It cannot indicate the sustainability of a nation's growth. It fails to account for the costs imposed on human health and the environment. Negative externalities arising from the production or consumption of the nation's output remain hidden. Technological progress and the quality of existing goods can hardly be indicated by the *GDP per capita*. Just take PCs, a PC today is a different product as it was 10 years ago it is difficult to take these differences into account. **Much more things could be mentioned here, see the talk of Robert Kennedy.**

   (b) (5 points) State which of the following are a part of the *GDP* and which are not part of the *GDP*.

   A) Social security payments received by a retired factory worker.
   ◯ Part of GDP √ **Not Part of GDP**

   B) Unpaid services of a family member in painting the family home.
   ◯ Part of GDP √ **Not Part of GDP**

   C) The income of a butcher.
   √ **Part of GDP** ◯ Not Part of GDP

   D) The monthly allowance a college student receives from home.
   ◯ Part of GDP √ **Not Part of GDP**

E) Rent received on an apartment.
√ **Part of GDP** ◯ Not Part of GDP

F) A €2 million increase in business inventories.
√ **Part of GDP** ◯ Not Part of GDP

G) Interest on a BMW bond.
√ **Part of GDP** ◯ Not Part of GDP

H) The purchase of an insurance policy.
√ **Part of GDP** ◯ Not Part of GDP

**Solution:**

A) Social security payments received by a retired factory worker. — **Excluded**. - A transfer payment from taxpayers for which no service is rendered (in this year).

B) Unpaid services of a family member in painting the family home. — **Excluded**. - Not a market transaction. If any payment is made, it will be within the family.

C) The income of a butcher. — **Included**. - Payment for a final service. You cannot pass on a tooth extraction!

D) The monthly allowance a college student receives from home. — **Excluded**. - A private transfer payment; simply a transfer of income from one private individual to another for which no transaction in the market occurs.

E) Rent received on an apartment. — **Included**. - Payment for the final service of housing.

F) A $2 billion increase in business inventories. — **Included**. - The increase in inventories could only occur as a result of increased production.

G) Interest on a BMW bond. — **Included.** Income received by the bondholder for the services derived by the corporation for the loan of money.

H) The purchase of an insurance policy. — **Included**. - Insurance is a final service. If bought by a household, it will be shown as consumption; if bought by a business, as investment—as a cost added to its real investment in physical capital.

2. **[Interest Rate Policy]**

(a) (6 points) Explain briefly four ways and channels, respectively, on how interest rate policy can have a positive effect on investments.

**Solution:** Basically, four ways exist how a reduced policy rate can spur investments:

1. Banks can lend more to households and companies, rather than holding on to

cash, which has now become costly.

2. Businesses can invest more, as funding investment is now cheaper.
3. Households could save less, or borrow to spend more.
4. Demand for the currency could fall. This might lead to a depreciation of the currency, an increase in the price of imported goods and growing demand for the country's exports which are now cheaper for foreign buyers.

(b) (4 points) Explain briefly why interest rate policy is less efficient in spurring investments when policy rates set by a central bank are already negative.

**Solution:**

*Limited Pass-Through to Deposit Rates:* Commercial banks have difficulties to charge costumers for taking their money. Low deposit rates comes along with a weak profitability of private banks which are likely to raise lending rates to save their interest margin.
*Limited Pass-Through to Lending Rates:* Commercial banks cannot pass-through it as lending rates become very low, more risky investments get financed which, in turn, brings economic uncertainties to the market.

3. **[Solow Model]**

Consider the Solow model of economic growth as introduced in the lecture. Assume the production function is

$$Y = K^{1/2}L^{1/2}$$

where $Y$ is output, $K$ is the capital stock, and $L$ is the labor force. The labor force (assumed equal to the population) grows at a constant rate $n$. The capital stock depreciates at a constant rate $\delta$. There is no exogenous technological progress ($g = 0.02$). The saving rate is $s$.

(a) (5 points) Show that the production function satisfies the assumption of the Solow model.

**Solution:**

If $K = 0$ or $L = 0$, $F(K, L) = 0$. Check constant returns to scale:

$$F(cK, cL) = (cK)^{\frac{1}{2}}(cL)^{\frac{1}{2}} = cK^{\frac{1}{2}}L^{\frac{1}{2}} \rightarrow \checkmark\text{CRS}$$

Check for positive but diminishing returns:

$$\frac{\partial F}{\partial L} = \frac{1}{2}K^{\frac{1}{2}}L^{-\frac{1}{2}} > 0 \quad \text{and} \quad \frac{\partial^2 F}{\partial L^2} = -\frac{1}{4}K^{\frac{1}{2}}L^{-\frac{3}{2}} < 0 \rightarrow \checkmark\text{positive and diminishing}$$

$$\frac{\partial F}{\partial K} = \frac{1}{2}K^{-\frac{1}{2}}L^{\frac{1}{2}} > 0 \quad \text{and} \quad \frac{\partial^2 F}{\partial K^2} = -\frac{1}{4}K^{-\frac{3}{2}}L^{\frac{1}{2}} < 0 \rightarrow \checkmark\text{positive and diminishing}$$

Alternatively, use directly the intensive form:

$$f'(k) = \frac{1}{2}k^{-\frac{1}{2}} > 0 \quad \text{and} \quad f''(k) = -\frac{1}{4}k^{-\frac{3}{2}} < 0 \rightarrow \checkmark\text{positive and diminishing}$$

(b) (3 points) Write the production function in the intensive form, i.e, in output per effective unit of labor terms.

**Solution:** $F(\frac{K}{L}, \frac{L}{L}) = \frac{Y}{L} = y = (\frac{K}{L})^{1/2}(\frac{L}{L})^{1/2} = k^{\frac{1}{2}}$

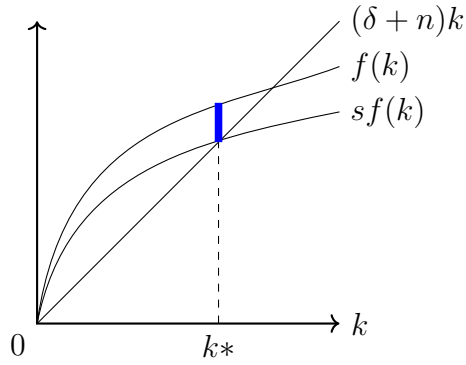
(c) (8 points) The dynamics of the capital stock per person are described by the equation

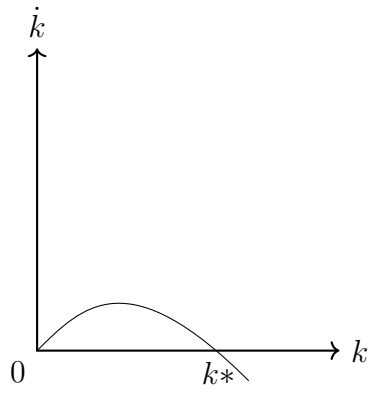
$$\dot{k} = sf(k) - (\delta + n)k.$$

Sketch how the steady-state stock of capital per worker is found using a $y$-$k$ diagram and explain why the economy will converge to this point in the long run. Moreover, sketch in the figure the overall output per worker function, $f(k)$, and show how much of the overall output per worker goes to consumption.

**Solution:**

The y-axis shows the investment per unit of effective labor.




$sf(k)$ is the proportion a country saves and hence invests. $(\delta - n)k$ refers to how much needs to be invested to keep the $k$ stable. Thus, the thick blue line denotes the fraction of overall production, $f(k)$, that goes to consumption. The dashed line is the proportion that is re-invested.

(d) (1 point) Can a country increase its speed of economic growth in the short run by an increase of its saving rate?

√ **YES**

◯ NO

(e) (1 point) Can a country increase its speed of economic growth in the long run by an increase of its saving rate?

√ **NO**

◯ YES

(f) (1 point) What is the speed of economic growth for the countries of this model in the long run.

**Solution:** Economic growth in a Solow growth model is determined only by growth in technology. In this model, however, technology plays no role thus there will be no growth in the long run. *(One of these two sentences would have been sufficent to get the point.)*

4. **[IS-LM Model]**

Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
Y &= C + I + G \\
C &= 300 + 0.25(Y - T) \\
I &= 400 + 0.25Y - 50i \\
G &= 200 \\
T &= 200 \\
\left(\frac{M}{P}\right)^d &= 2Y - 200i \\
\frac{M}{P} &= 600
\end{aligned}
$$

(a) (8 points) Derive the IS relation.

**Solution:**

$$
\begin{aligned}
Y &= 300 + 0.25(Y - 200) + 400 + 0.25Y - 50i + 200 \\
Y &= 850 + \frac{1}{2}Y - 50i \\
Y &= 1700 - 100i \\
i &= 17 - \frac{1}{100}Y
\end{aligned}
$$

(b) (3 points) Derive the LM relation.

**Solution:**

$$
\begin{aligned}
600 &= 2Y - 200i \\
Y &= 300 + 100i \\
i &= \frac{1}{100}Y - 3
\end{aligned}
$$

(c) (4 points) Solve for equilibrium output, $Y^*$, and interest rate, $i^*$.

**Solution:**

$$17 - \frac{1}{100}Y = \frac{1}{100}Y - 3$$

$$\frac{2}{100}Y = 20$$
$$Y^* = 1000$$

Plugging that into either the LM or IS relation, we get:

$$\frac{1}{100} \cdot 1000 - 3 = 7 = i^*$$

(d) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if government would –all else equal– reduce the tax?

- √ $i^*$ **would increase**
- ◯ $i^*$ would decrease
- √ $Y^*$ **would increase**
- ◯ $Y^*$ would decrease

(e) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if government would –all else equal– reduce the government spendings $G$?

- ◯ $i^*$ would increase
- √ $i^*$ **would decrease**
- ◯ $Y^*$ would increase
- √ $Y^*$ **would decrease**

(f) (2 points) What would happen to the equilibrium real output, $Y^*$, and interest rate, $i^*$, if the central bank would –all else equal– increase money supply, $\frac{M}{P}$?

- ◯ $i^*$ would increase
- √ $i^*$ **would decrease**
- √ $Y^*$ **would increase**
- ◯ $Y^*$ would decrease

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This box is for the examiner only.

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total |
|---|---|---|---|---|---|---|---|---|
| Points: | 6 | 8 | 21 | 6 | 21 | 16 | 12 | 90 |
| Score: | | | | | | | | |

1. (6 points) **[Balance of payments]** Explain why net exports must equal net capital outflows using the *formal representation*.

2. (8 points) **[Unemployment]** There are several rational explanations why wages are downward rigid and remain above the market clearing level. Explain two of the sources we discussed in class.

3. **[IS-LM Model: Equilibrium]** Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 400 + 0.2(Y - T) \\
I &= 200 + 0.4Y - 1000i \\
G &= 150 \\
T &= 200 \\
\left(\frac{M}{P}\right)^d &= 4Y - 8000i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 1700 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

Given the information above, please answer the following questions:

(a) (7 points) Derive the IS curve.

(b) (4 points) Derive the LM curve.

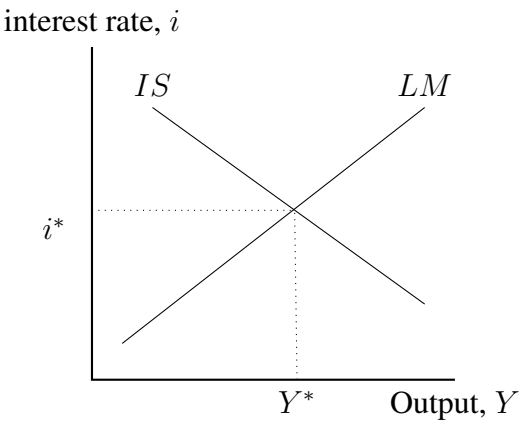
(c) (4 points) Solve for the equilibrium output, $Y^*$.

(d) (2 points) Solve for the equilibrium interest rate, $i^*$.

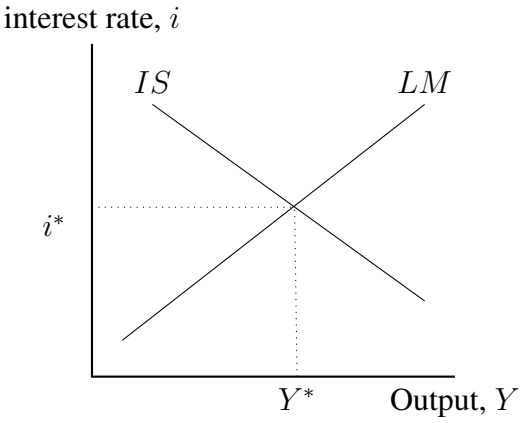
(e) (4 points) Solve for the equilibrium consumption, $C^*$, and investment $I^*$.

4. **[IS-LM Model: The effects of fiscal and monetary policy]**

(a) (3 points) Briefly sketch and discuss the implications of monetary expansion in the IS-LM model using the diagram below.



(b) (3 points) Briefly sketch and discuss the impact of an increase in taxes in the IS-LM model using the following diagram.




5. **[Measuring the Economy]**

(a) (8 points) Assume that two countries have the same nominal GDP (measured in the same currency and according to the same accounting rules). Explain why you cannot assume that citizens in both countries enjoy roughly the same level of economic well-being.

(b) (4 points) Discuss why it is not correct to estimate GDP by adding the value of all goods and services produced, both final goods and intermediate goods.

(c) (4 points) The average workweek has declined over the past 70 years and is currently about 35 hours a week. Discuss how this change has affected real GDP **and** well-being.

(d) (5 points) Explain briefly the five steps to calculate the Consumer Price Index.

6. **[Solow Model]**

Consider the Solow model of economic growth as introduced in the lecture. Assume the production function is

$$Y = K^{1/2}L^{1/2}$$

where $Y$ is output, $K$ is the capital stock, and $L$ is the labor force. The labor force (assumed equal to the population) grows at a constant rate $n$. The capital stock depreciates at a constant rate $\delta$. There is no exogenous technological progress $(g = 0)$. The saving rate is $s$.

(a) (5 points) Show that the production function satisfies the assumption of the Solow model.

(b) (3 points) Write the production function in the intensive form, i.e, in output per effective unit of labor terms.

(c) (8 points) The dynamics of the capital stock per person are described by the equation

$$\dot{k} = sf(k) - (\delta + n)k.$$

Sketch how the steady-state stock of capital per worker is found using a $y$-$k$ diagram and explain why the economy will converge to this point in the long run. Moreover, sketch in the figure the overall output per worker function, $f(k)$, and show how much of the overall output per worker goes to consumption.

7. **[Interest Rate Policy]**

   (a) (2 points) State the primary objective of the European Central Bank.

   (b) (6 points) Explain briefly three ways and channels, respectively, on how interest rate policy can have a positive effect on investments.

   (c) (4 points) Explain briefly why interest rate policy is less efficient in spurring investments when policy rates set by a central bank are already negative.

*End of exam*

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**This box is for the examiner only.**

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 9 | 6 | 26 | 8 | 12 | 29 | 90 |
| Score: | | | | | | | |

1. (9 points) **[Unemployment]** Explain three rational explanations why wages are downward rigid and remain above the market clearing level.

2. (6 points) **[Balance of payments]**

   The equation
   $$Y = C + I + G + EX - IM$$
   is often used to describe an open economy. Use this representation to explain why net exports must equal net capital outflows.

3. **[Inflation]**

   (a) (5 points) Explain how the inflation rate can be computed with the *consumer price index* (CPI) in five steps.

   (b) (6 points) Briefly explain three ways or channels through which the European Central Bank can have a negative impact on investment by raising policy rates.

   > **Solution:** Basically, four ways exist how a reduced policy rate can spur investments:
   >
   > 1. Banks can lend more to households and companies, rather than holding on to cash, which has now become costly.
   > 2. Businesses can invest more, as funding investment is now cheaper.
   > 3. Households could save less, or borrow to spend more.
   > 4. Demand for the currency could fall. This might lead to a depreciation of the currency, an increase in the price of imported goods and growing demand for the country's exports which are now cheaper for foreign buyers.

   (c) (7 points) Explain why central banks seek to keep inflation low. Discuss arguments why high inflation is bad for the economy and society.

   (d) (8 points) Explain the *Taylor rule*. In particular, write down and explain the formal expression of the rule.

4. (8 points) **[Reverse multiplier effect]** Explain what we meant in lecture by the phrase *reverse multiplier effect*. Discuss what governments or central banks can do to fight or mitigate the reverse multiplier effect.

5. (12 points) **[Measuring the Economy]**

   Explain what the GDP measures and discuss why it can only be considered as an imperfect measurement of well-being.

6. **[IS-LM Model: Equilibrium]** Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 600 + 0.2(Y - T) \\
I &= 300 + 0.4Y - 1000i \\
G &= 200 \\
T &= 100 \\
\left(\frac{M}{P}\right)^d &= 5Y - 5000i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 1600 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

Given the information above, please answer the following questions:

(a) (7 points) Derive the IS curve.

(b) (4 points) Derive the LM curve.

(c) (4 points) Solve for the equilibrium output, $Y^*$.

(d) (2 points) Solve for the equilibrium interest rate, $i^*$.

(e) (4 points) Solve for the equilibrium consumption, $C^*$, and investment $I^*$.

(f) (8 points) Assume government decides to increase aggregate demand by increasing government spending by 21 so that $G = 221$. Show the implications of that expansive fiscal policy for the IS and the LM curve. Sketch the changes in a plot with the interest rate, $i$, and the Oputput, $Y$, on the axes. Explain what will happen to the equilibrium. Will aggregate output increase by more or less than 21? Justify your answer.

# References

*End of exam*

23-01_macro.pdf

This box is for the examiner only.

| Question: | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Points: | 30 | 10 | 20 | 18 | 12 | 90 |
| Score: | | | | | | |

1. **[IS-LM Model: Equilibrium]** Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 80 + 0.8(Y - T) \\
I &= 150 - 1000i \\
G &= 100 \\
T &= 100 \\
\left(\frac{M}{P}\right)^d &= 0.02Y - 1000i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 50 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

Given the information above, please answer the following questions:

(a) (10 points) Derive the IS and the LM relation.

(b) (10 points) Solve for

- the equilibrium output, $Y^*$,
- the equilibrium interest rate, $i^*$,
- the equilibrium consumption, $C^*$, and
- investment $I^*$.

(c) (10 points) Assume government decides to increase aggregate demand by increasing government spending so that $G' = 200$.

- Sketch the implications of that expansive fiscal policy for the IS and the LM curve in a plot with the interest rate, $i$, and the output, $Y$, on the axes.
- Calculate
  - the equilibrium output, $Y'^*$,
  - the equilibrium interest rate, $i'^*$,
  - the equilibrium consumption, $C'^*$, and
  - investment $I'^*$.

2. (10 points) **[Keynes vs. Hayek]** Explain the Keynesian strategy to fight economic bust and boom cycles and discuss how the Austrian school including Friedrich Hayek think economies should deal with these cycles.

3. (20 points) **[Comparing the financial crisis to the Corona crisis]**

Explain how the financial crisis of 2008/2009 differs from the Corona crisis. In particular, discuss gross domestic product, private consumption, the savings rate, government spending, corporate insolvencies, unemployment and the development of price levels.

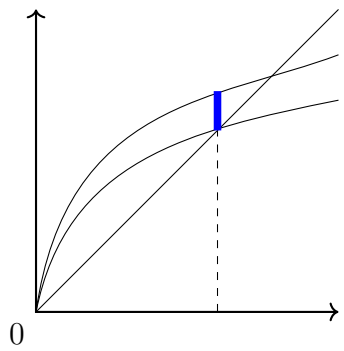
4. (18 points) **[Solow]**

   a) Discuss ways to boost economic growth under the Solow model.

   b) Explain what is meant by the term "steady-state equilibrium" in the Solow model.

   c) Explain the assumptions that guarantee the stability of the economic growth path in the Solow model.

   d) Label the axes and the functions of the figure below. Explain the three functions and the meaning of the blue and dashed lines.




5. (12 points) **[Measuring the Economy]** Explain the difficulties in measuring the consumer price index over time and discuss why it can only be considered an imperfect measure of the cost of living.

*End of exam*

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| Question: | 1 | 2 | 3 | Total |
|---|---|---|---|---|
| Points: | 41 | 34 | 15 | 90 |
| Score: | | | | |

1. **[Various]**

   (a) (4 points) Explain the difference of a positive and a normative statement in economics.

   (b) (5 points) Identify and explain plausible reasons why macroeconomists hold divergent views.

   (c) (6 points) The GDP is often used to indicate living standards. Identify and explain briefly alternative measures that can be used to evaluate the average living standards.

   (d) (6 points) Provide an explanation of the *simple rule for r* in international economics.

   (e) (6 points) Elaborate on the meaning and derivation of the equation $NCO = NEX$ as discussed in the lecture.

   (f) (6 points) Explain the concept of the "magic of the price system" as presented by Milton Friedman.

   (g) (8 points) Describe the Keynesian approach to managing economic bust and boom cycles. Moreover, discuss the potential long-term negative consequences of implementing this strategy.

2. **[IS-LM Model: Equilibrium]** Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 100 + 0.25(Y - T) \\
I &= 300 + 0.25Y - 50i \\
G &= 50 \\
T &= 10 \\
\left(\frac{M}{P}\right)^d &= 3Y - 200i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 600 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

   Given the information above, please answer the following questions:

   (a) (10 points) Derive the IS and the LM relation.

   (b) (12 points) Solve for

   - the equilibrium output, $Y^*$,
   - the equilibrium interest rate, $i^*$,
   - the equilibrium consumption, $C^*$, and
   - investment $I^*$.

   (c) (12 points) One of the key concepts covered in this course was the IS-LM model and its applications to fiscal and monetary policy. As part of your preparation for the exam, you should be familiar with the table titled "The effects of fiscal and monetary policy

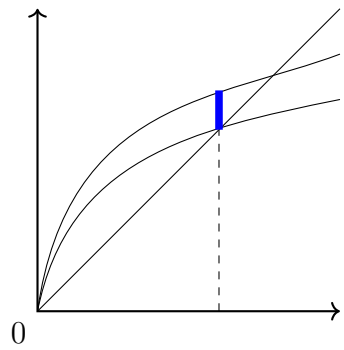
in the IS-LM model" found in the lecture notes. This table outlines the directional effects of various policy changes on key economic variables in the model. Please fill the gaps of the following table using the terms *left*, *right*, *none*, *down*, and *up*.

| | Shift of IS | Shift of LM | Movement in Output | Movement in Interest Rate |
|---|---|---|---|---|
| Increase in taxes | | | | |
| Decrease in taxes | | | | |
| Increase in spending | | | | |
| Decrease in spending | | | | |
| Increase in money | | | | |
| Decrease in money | | | | |

3. (15 points) **[Solow]**

   (a) Describe the key insights of the Solow Model and the policy implications of the Solow Model.

   (b) Label the axes and the functions of the figure below. Explain the three functions and the meaning of the blue and dashed lines.




*End of exam*

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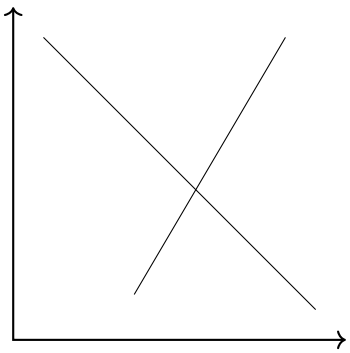
This box is for the examiner only.

| Question: | 1 | 2 | 3 | Total |
|---|---|---|---|---|
| Points: | 35 | 31 | 24 | 90 |
| Score: | | | | |

1. (35 points) **[Unemployment]**

   Economists frequently employ a plot featuring two linear functions to depict the market equilibrium of full employment and unemployment, respectively.

   (a) Explain the concept of unemployment using the provided sketch. Label the axes and functions accurately and illustrate the situation in which the market fails to clear. Additionally, discuss the three rational explanations for the existence of unemployment that have been discussed in class.



   (b) Unemployment poses significant economic and social challenges at both the individual and societal levels. Discuss the individual costs associated with unemployment, as well as the economic and social costs that arise from high levels of unemployment.

   (c) Describe briefly the proposition of Okun's Law.

   (d) The *reverse multiplier effect* resulted in high unemployment levels during the Great Depression, which began in August 1929 after the economic expansion of the *Roaring Twenties* came to an abrupt halt. Explain the reverse multiplier effect briefly and discuss policy actions that can mitigate the effect.

2. **[Measuring the economy]**

   (a) (9 points) Provide an explanation of GDP, including how it is measured. Additionally, discuss briefly why GDP is just an imperfect welfare measure.

   (b) (12 points) Consider the following table of price and quantities produced for a small economy. Assume that only four goods (A, B, C) have been produced in the economy. B is an intermediate good that is used in the production of good C.

| Year | A | | B | | C | |
|---|---|---|---|---|---|---|
| | Price | Quantity | Price | Quantity | Price | Quantity |
| 2020 | €1 | 2000 | €2 | 100 | €10 | 10 |
| 2021 | €1 | 1800 | €3 | 50 | €10 | 20 |

   a) Calculate the nominal GDP for 2020 and 2021.

   b) Calculate the percentage change in nominal GDP between 2020 and 2021.

c) Calculate the consumer price index (CPI) in 2020 and 2021, using 2020 as the base year.

d) Calculate the real GDP for 2020 and 2021.

(c) (10 points) The consumer price index (CPI) is an accurate measure of the selected goods that aim to be a representative bundle of goods, but it is not a perfect measure of the cost of living. Why? Explain briefly the three sources of bias which we discussed in class.

3. **[IS-LM Model: Equilibrium]**

(a) (12 points) Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 10 + 0.8(Y - T) \\
I &= 1000 + 0.1Y - 0.05i \\
G &= 16 \\
T &= 4 \\
\left(\frac{M}{P}\right)^d &= 20Y - 2i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 50 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

Given the information above, derive the IS relation and the LM relation. Then, solve for the equilibrium output, $Y^*$, the equilibrium interest rate, $i^*$, the equilibrium consumption, $C^*$, and investment $I^*$.

(b) (12 points) One of the key concepts covered in this course was the IS-LM model and its applications to fiscal and monetary policy. As part of your preparation for the exam, you should be familiar with the table titled "The effects of fiscal and monetary policy in the IS-LM model" found in the lecture notes. This table outlines the directional effects of various policy changes on key economic variables in the model. Please fill the gaps of the following table using the terms *left*, *right*, *none*, *down*, and *up*.

| | Shift of IS | Shift of LM | Movement in Output | Movement in Interest Rate |
|---|---|---|---|---|
| Increase in taxes | | | | |
| Decrease in taxes | | | | |
| Increase in spending | | | | |
| Decrease in spending | | | | |
| Increase in money | | | | |
| Decrease in money | | | | |

*End of exam*

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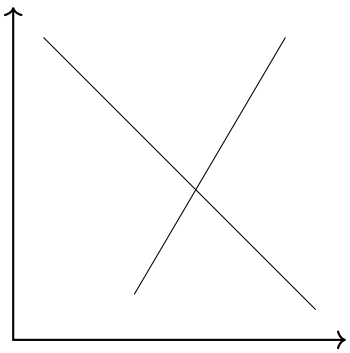
**This box is for the examiner only.**

| Question: | 1 | 2 | 3 | 4 | 5 | 6 | 7 | Total |
|---|---|---|---|---|---|---|---|---|
| Points: | 35 | 6 | 10 | 6 | 12 | 15 | 6 | 90 |
| Score: | | | | | | | | |

1. (35 points) **[Unemployment]**

   Economists frequently employ a plot featuring two linear functions to depict the market equilibrium of full employment and unemployment, respectively.

   (a) Explain the concept of unemployment using the provided sketch. Label the axes and functions accurately and illustrate the situation in which the market fails to clear. Additionally, discuss the three rational explanations for the existence of unemployment that have been discussed in class.



   (b) Unemployment poses significant economic and social challenges at both the individual and societal levels. Discuss the individual costs associated with unemployment, as well as the economic and social costs that arise from high levels of unemployment.

   (c) Describe briefly the proposition of Okun's Law.

   (d) The *reverse multiplier effect* resulted in high unemployment levels during the Great Depression, which began in August 1929 after the economic expansion of the *Roaring Twenties* came to an abrupt halt. Explain the reverse multiplier effect briefly and discuss policy actions that can mitigate the effect.

2. (6 points) **[GDP]**

   The GDP per capita is a monetary measure of the market value of all the final goods and services produced in one year. Please elaborate on why nominal GDP per capita is an imperfect or unsuitable measure for conducting cross-country studies on economic prosperity.

3. (10 points) **[Consumer price index]** The consumer price index (CPI) is an accurate measure of the selected goods that aim to be a representative bundle of goods, but it is not a perfect measure of the cost of living. Why? Explain briefly the three sources of bias which we discussed in class.

4. (6 points) **[Balance of payments]** Explain briefly why net exports must equal net capital outflows using the *formal representation*.

5. (12 points) **[IS-LM Model: Equilibrium]**

   Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 100 + 0.2(Y - T) \\
I &= 300 + 0.1Y - 60i \\
G &= 90 \\
T &= 500 \\
\left(\frac{M}{P}\right)^d &= 20Y - 2i \qquad \rightarrow \text{[real money demand]} \\
\left(\frac{M}{P}\right)^s &= 50 \qquad \rightarrow \text{[real money supply]}
\end{aligned}
$$

   Given the information above, derive the IS relation and the LM relation. Then, solve for the equilibrium output, $Y^*$, the equilibrium interest rate, $i^*$, the equilibrium consumption, $C^*$, and investment $I^*$.

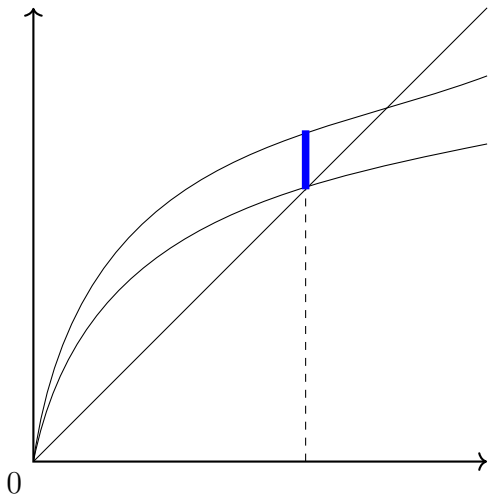
6. (15 points) **[Solow model]**

   a) Discuss ways to boost economic growth under the Solow model.

   b) Explain what is meant by the term *steady-state equilibrium* in the Solow model.

   c) Explain what is meant with the *golden rule of consumption* and at what point of the output function is consumption maximized.

   d) Label the axes and the functions of the figure below. Explain the three functions and the meaning of the blue and dashed lines.



7. (6 points) **[Exchange rates]**

Suppose an investor from Indonesia asks for your help. The Indonesian investor wants to buy new inputs for his production in Indonesia next year, i.e., t=2024. Today, i.e., t=2023, she has 10,000,000 Indonesian Rupiah (IDR) and she wonders what to do with this money for one year. She has two potential investments in mind. One is an investement in Europe with 4% of annual interest and the other one is an investment in Indonesia that offers 16% of annual interest. Where should she invest given the following conditions:

- One Euro can be converted to 15,000 IDR this year.
- You expect that one Euro can be converted to 20,000 IDR next year.
- Moreover, you expect no inflation in both countries and no banking fees or alike.

Please complete the sentences:

- The value of your investment in Indonesia in the year 2024 is

  _____ IDR.

- The value of your investment in Europe in the year 2024 is

  _____ IDR.

*End of exam*

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This box is for the examiner only.

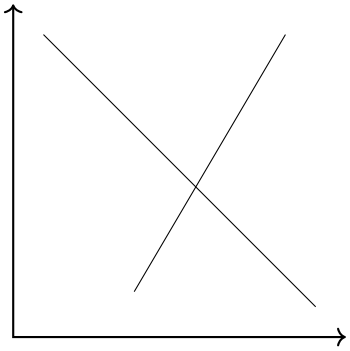
| Question: | 1 | 2 | 3 | 4 | 5 | 6 | Total |
|---|---|---|---|---|---|---|---|
| Points: | 15 | 27 | 6 | 10 | 17 | 15 | 90 |
| Score: | | | | | | | |

# Macroeconomics (01/2025)

1. **[Unemployment]**

   Economists frequently employ a plot featuring two linear functions to depict the market equilibrium of full employment and unemployment, respectively.

   (a) (6 points) Explain briefly the concept of unemployment using the provided sketch. Label the axes and functions accurately and illustrate the situation in which the market fails to clear.



   (b) (9 points) Explain briefly the three rational explanations for the existence of unemployment that have been discussed in class.

2. **[IS-LM Model]**

   (a) (4 points) Explain what was referred to in the lecture as the *reverse multiplier effect*.

   (b) (10 points) Discuss measures that governments or central banks can implement to counteract or mitigate the *reverse multiplier effect*. In particular, refer to the IS-LM Model and identify the variables that can increase aggregate demand. For each variable, explain the necessary direction of change to boost aggregate demand.

   (c) (5 points) Briefly explain what is meant by the term *multiplier effect* in the context of the IS-LM model.

   (d) (8 points) Explain the Keynesian strategy and the Austrian school strategy for addressing economic boom and bust cycles. Discuss arguments that warn against excessive use of Keynesian policies due to their potentially adverse long-term consequences.

3. (6 points) **[Balance of payments]**

   In the lecture, we learned that "Net exports must equal net capital outflow". Briefly verify and explain this claim using the formal representation of an open economy.

4. (10 points) **[Inflation]**

   Central banks aim to keep inflation low because they are concerned about the negative effects that high inflation rates can have on an economy. Explain the specific negative effects

that central banks seek to avoid.

5. **[Measuring the Economy]**

Suppose an economy consists of only three goods: Good A, B, and C. Good A, Good B, and Good D are final goods and Good C is an intermediate good. The sales and price data for these goods over two different years are provided below:

| Good | A | | B | | C | | D | |
|---|---|---|---|---|---|---|---|---|
| Year | Price | Quantity | Price | Quantity | Price | Quantity | Price | Quantity |
| 2023 | €3 | 150 | €3 | 80 | €1 | 20 | €4 | 10 |
| 2024 | €3 | 120 | €5 | 140 | €2 | 16 | €6 | 8 |

(a) (6 points) Calculate the nominal GDP for both years.

(b) (4 points) Calculate the real GDP for both years using 2023 as the base year.

(c) (4 points) Calculate the Consumer Price Index for both years using 2023 as the base year.

(d) (3 points) What was the inflation rate in this economy between 2019 and 2020? Express your answer as a percentage.

6. (15 points) **[Elective task: Choose Task A or Task B]**

Please choose **one** out of the following **two** tasks to work on. If you complete more than one tasks, only task A will be evaluated. If you have started working on an task but have changed your mind, please cross out the task.

## Task A: Fundamental causes of income difference

In the Handbook article by Acemoglu et al. (2005) which we discussed in class, three *fundamental causes of income differences* are explained. Name and describe these three causes, providing a detailed explanation for each.

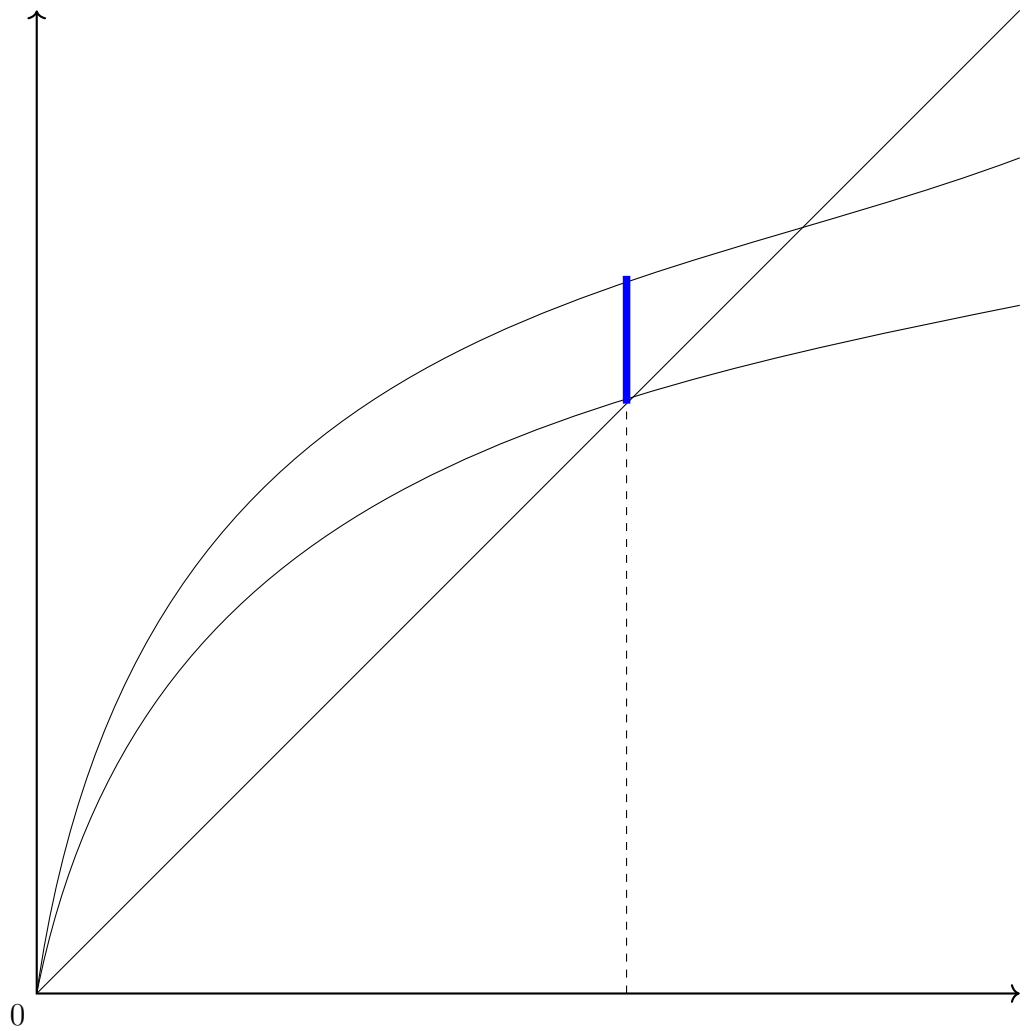
## Task B: Solow model

a) Discuss ways to boost economic growth under the Solow model.

b) Explain what is meant by the term *steady-state equilibrium* in the Solow model.

c) Label the axes and the functions of the figure below. Explain the three functions and the meaning of the blue and dashed lines.

d) Explain what is meant with the *golden rule of consumption* and at what point of the output function is consumption maximized.




# References

Acemoglu, D., Johnson, S., and Robinson, J. A. (2005). Institutions as a fundamental cause of long-run growth. In Aghion, P. and Durlauf, S. N., editors, Handbook of Economic Growth, volume 1A, chapter 6, pages 385–472. Elsevier.

*End of exam*

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This box is for the examiner only:

| Question: | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Points: | 27 | 24 | 12 | 12 | 15 | 90 |
| Score: | | | | | | |

# Macroeconomics (04/2025)

1. **Various**

   (a) (14 points) In the lecture, we defined seven assumptions of perfect markets and perfect competition. Name and explain these assumptions.

   (b) (4 points) In the lecture, we discussed the two fundamental theorems of welfare economics. Explain these two theorems.

   (c) (9 points) Unemployment is a form of market failure. Provide three rational explanations for why labor markets may not clear, resulting in unemployment.

2. **GDP and CPI**

   Suppose an economy produces three goods: A, B, and C. Goods A and B are final goods. Good C is an intermediate good. The following table shows the quantities and prices for two years. The prices are measured in €.

| Year | A (quantity) | Price per good A | B (quantity) | Price per good B | C (quantity) | Price per good C |
|---|---|---|---|---|---|---|
| 2022 | 1,000 | 2 | 500 | 1 | 200 | 3 |
| 2023 | 1,200 | 3 | 600 | 2 | 300 | 4 |

   (a) (2 points) Calculate nominal GDP for both years.

   (b) (2 points) Using 2022 as the base year, calculate real GDP for both years.

   (c) (2 points) Using 2022 as the base year, calculate Consumer Price Index (CPI) for both years.

   (d) (2 points) Calculate the inflation rate from 2022 to 2023

   (e) (4 points) Discuss why nominal GDP might not be a good measure of economic well-being.

   (f) (4 points) Central banks and politicians try to keep inflation low because it has negative implications for an economy. Discuss four negative implications of high inflation.

   (g) (8 points) Discuss two alternative measures to GDP that can provide insights into national welfare. In particular, elaborate on what these measures do better and worse than GDP.

3. (12 points) **IS-LM Model**

Consider an economy described by the following equations of the IS-LM model as discussed in the lecture:

$$
\begin{aligned}
C &= 110 + 0.3(Y - T) \\
I &= 270 + 0.2Y - 50i \\
G &= 35 \\
T &= 10 \\
\left(\frac{M}{P}\right)^d &= 3Y - 200i \qquad \rightarrow \text{real money demand} \\
\left(\frac{M}{P}\right)^s &= 600 \qquad \rightarrow \text{real money supply}
\end{aligned}
$$

Given the information above, derive the IS relation and the LM relation. Then, solve for the equilibrium output, $Y^*$, the equilibrium interest rate, $i^*$, the equilibrium consumption, $C^*$, and investment $I^*$.

4. (12 points) **Fundamental causes of income difference**

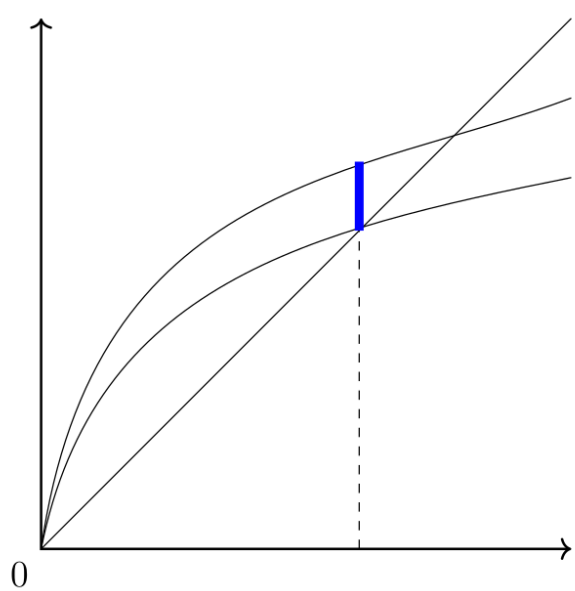
   In the Handbook article by Acemoglu et al. (2005) which we discussed in class, three *fundamental causes of income differences* are explained. Name and describe these three causes.

5. (15 points) **Solow model**

   a) Discuss ways to boost economic growth under the Solow model.

   b) Explain what is meant by the term *steady-state equilibrium* in the Solow model.

   c) Label the axes and the functions of the figure below. Explain the three functions and the meaning of the blue and dashed lines.



d) Explain what is meant by the *golden rule of consumption* and at what point of the output function consumption is maximized.

# References

Acemoglu, D., Johnson, S., & Robinson, J. A. (2005). Institutions as a fundamental cause of long-run growth. In P. Aghion & S. N. Durlauf (Eds.), *Handbook of economic growth* (Vol. 1A, pp. 385–472). Elsevier.

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This box is for the examiner only:

| Question: | 1 | 2 | 3 | 4 | 5 | Total |
|---|---|---|---|---|---|---|
| Points: | 6 | 4 | 8 | 6 | 21 | 45 |
| Score: | | | | | | |

# Macroeconomics (07/2025)

1. (6 points) **Ricardo**

   Assume that only two countries, A and B, exist. Both countries are equally endowed with labor which is the only production factor. Both countries can produce either good $y$ or good $x$. The table below gives the input coefficients, $a$, for both countries, i.e., the units of labor needed to produce one unit of good $y$ and good $x$, respectively. Assume that both countries have 12 units of labor available.

| | $x$ | $y$ |
|---|---|---|
| $B$ | 20 | 25 |
| $A$ | 10 | 13 |

   a) Specify the country with an **absolute** advantage in good $x$. Explain your decision.
   b) Specify the country with an **absolute** advantage in good $y$. Explain your decision.
   c) Specify the country with a **comparative** advantage in good $x$. Explain your decision.
   d) Specify the country with a **comparative** advantage in good $y$. Explain your decision.

2. (4 points) **Heckscher-Ohlin model**

   Given the assumptions of the Heckscher-Ohlin Model, let's consider a scenario involving only two countries, A and B, and two goods, $x$ and $y$. It is assumed that good $y$ is capital-intensive in its production, whereas good $x$ is labor-intensive. Given the endowments in the table below, name the comparative advantages of the two countries. Briefly discuss your decision.

| | Countries | |
|---|---|---|
| | A | B |
| Labor Endowments | 7000 | 20 |
| Capital Endowments | 9500 | 33 |

3. (8 points) **Solow model**

   Briefly explain what the Solow model investigates, and discuss ways to boost economic growth in the short and long run within the Solow model. Explain what the term steady-state growth rate means in the Solow model.

4. (6 points) **WTO**

   Describe the basic principles of the World Trade Organization (WTO) that we have discussed in the lecture notes.

5. **Monetary international economics**

   (a) (8 points) One of Donald Trump's economic goals is to reduce the United States trade deficit. Using the formal representation, explain how he can achieve this.

   (b) (7 points) Since Donald Trump returned to the Oval Office, the Dollar-to-Euro exchange rate has changed significantly. Has the Dollar appreciated or depreciated against the Euro? Discuss how this change may affect the United States' trade deficit.

   (c) (6 points) Suppose you aim to invest 1000 Euro from 2025 to 2026 in Switzerland. Calculate the value of the investment in the year 2026 given the following conditions:

   - The annual interest rate in is 2%.
   - 0.94 Swiss Franc can be converted to 1 Euro in 2025.
   - You expect that 1 Euro will cost 0.02 Swiss Franc less in 2026.
   - Moreover, you expect no inflation in the Swizerland.

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This box is for the examiner only:

| Question: | 1 | 2 | 3 | 4 | Total |
|---|---|---|---|---|---|
| Points: | 11 | 4 | 8 | 22 | 45 |
| Score: | | | | | |

# Macroeconomics (11/2025)

1. (11 points) **Ricardo**

   Assume that only two countries, A and B, exist. Both countries are equally endowed with labor which is the only production factor. Both countries can produce either good $y$ or good $x$. The table below gives the input coefficients, $a$, for both countries, i.e., the units of labor needed to produce one unit of good $y$ and good $x$, respectively.

| | $x$ | $y$ |
|---|---|---|
| $B$ | 20 | 70 |
| $A$ | 40 | 140 |

   a) Specify the country with an **absolute** advantage in good $x$. Explain your decision.
   b) Specify the country with an **absolute** advantage in good $y$. Explain your decision.
   c) Specify the country with a **comparative** advantage in good $x$. Explain your decision.
   d) Specify the country with a **comparative** advantage in good $y$. Explain your decision.
   e) Assume the input coefficient of country $A$ in good $Y$ changes to $a_A^y = 139$. All other input coefficients remain the same. Discuss the trade structure in that situation.

2. **Heckscher-Ohlin model**

   (a) (4 points) Given the assumptions of the Heckscher-Ohlin model, let's consider a scenario involving only two countries, A and B, and two goods, $x$ and $y$. It is assumed that good $y$ is capital-intensive in its production, whereas good $x$ is labor-intensive. Given the endowments in the table below, name the comparative advantages of the two countries. Briefly discuss your decision.

| | Countries | |
|---|---|---|
| | A | B |
| Labor Endowments | 10 | 20 |
| Capital Endowments | 14 | 24 |

3. (8 points) **Solow model**

   Briefly explain what the Solow model investigates, and discuss ways to boost economic growth in the short and long run within the Solow model. Explain what the term steady-state growth rate means in the Solow model.

4. **Monetary international economics**

   (a) (7 points) Describe the *purchasing power parity assumption* as it was explained in the lecture. In particular, name the conditions for the assumption to hold.

(b) (9 points) The article "TRUMP EFFECT: A Running List of New U.S. Investment in President Trump's Second Term" by The White House (2025), published on August 15, contains a frequently updated list of new U.S.-based investments in President Trump's second term. Here is a quote from the article:

> "Since President Donald J. Trump took office, his unwavering commitment to revitalizing American industry has spurred trillions of dollars of investments in U.S. manufacturing, production, and innovation — and the list only continues to grow."

Comment on the quote and discuss the potential consequences of this policy for the U.S. trade balance. Justify your argument using formal economic representation.

(c) (6 points) Suppose you aim to invest 1000 Euro from 2025 to 2026 in China. Calculate the value of the investment in Euro in the year 2026 given the following conditions:

- The annual interest rate in is 8%.
- 1 Renminbi can be converted to 12 Eurocent in 2025.
- You expect that the Renminbi will cost 10% more Euro in 2026.
- Moreover, you expect no inflation in both countries.

# References

The White House. (2025). *TRUMP EFFECT: A running list of new U.S. Investment in President Trump's second term*. https://www.whitehouse.gov/articles/2025/08/trump-effect-a-running-list-of-new-u-s-investment-in-president-trumps-second-term/